Human Activity Recognition Using LSTM

A deep learning project that classifies everyday activities from smartphone sensor data.



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Why activity recognition matters

Recognize the activity

Human Activity Recognition identifies what a person is doing from sensor measurements.

Make sensing accessible

Smartphone accelerometers and gyroscopes make activity sensing accessible for real-world applications.

Classify everyday movement

Project focus: classify walking, stair movement, sitting, standing, and laying.

Project focus

From sensor measurements to a predicted activity label.

Smartphone accelerometers · gyroscopes

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The project goal is an end-to-end classification pipeline

01

Load and inspect

Load and inspect the HAR dataset.

02

Prepare inputs

Prepare labels and numeric features.

03

Train and validate

Train and validate a multi-class LSTM model.

04

Evaluate and export

Evaluate performance and export reusable model artifacts.

01 → 04

Each stage prepares the output needed by the next.

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The dataset captures everyday movement through smartphone sensors

Sensor measurements

Measurements come from smartphone accelerometers and gyroscopes while participants perform daily activities.

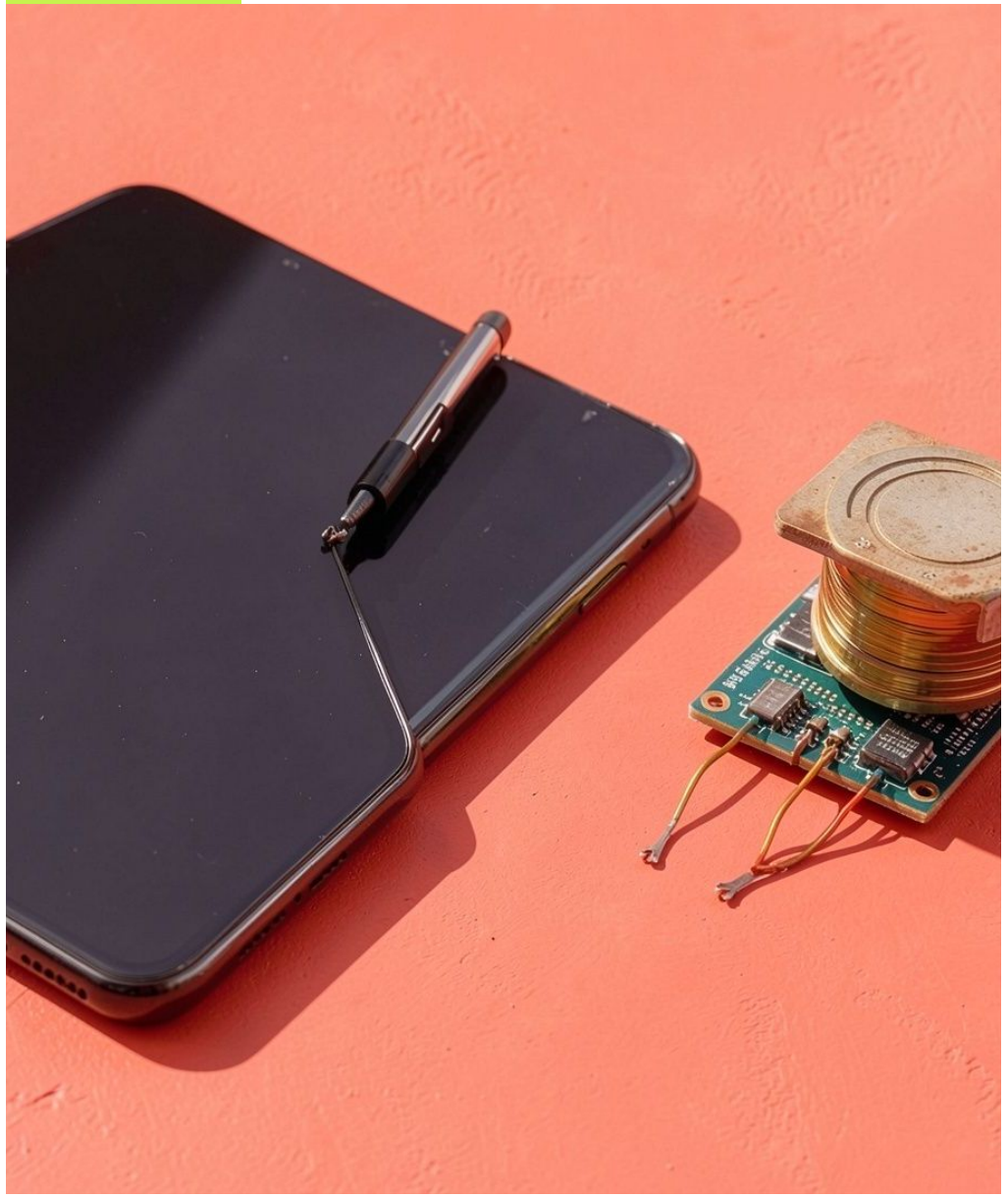
Activity labels

Activity labels are encoded for multi-class neural-network training.

Version check

Exact file names and label names should be confirmed from the final dataset version.

DATA INPUT



Smartphone sensors provide the measurements; encoded activities provide the targets.

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Preprocessing converts raw features into model-ready sequences

01

Inspect

Inspect dimensions, columns, missing values, and target labels.

02

Align

Retain numeric features and align them with the correct labels.

03

Split

Use a stratified 80/20 training-validation split.

04

Scale and reshape

Fit StandardScaler on training data, transform validation data, then reshape inputs to (samples, features, 1).

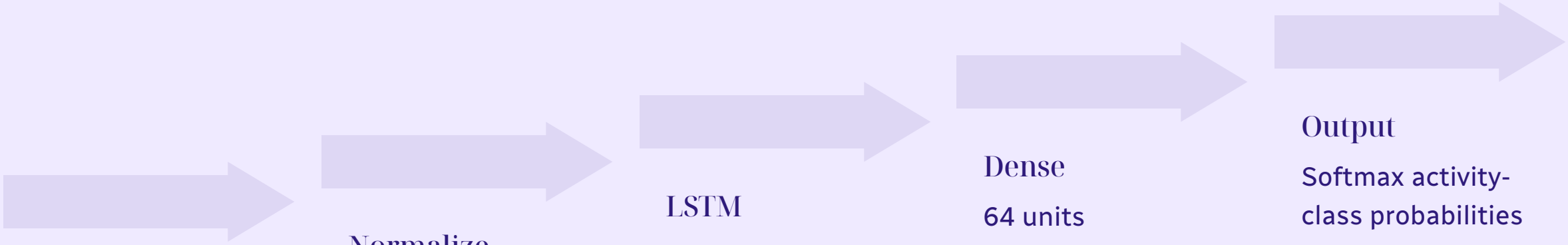
Data contract

The validation set stays separate while the scaler learns only from training data.

80 / 20 stratified split

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The LSTM architecture learns a compact activity representation



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Training balances learning speed and generalization

Optimizer

Adam

Loss

categorical cross-entropy

Maximum training schedule

up to 30 epochs, batch size 64

Early stopping and learning-rate reduction

help respond when validation progress stalls.

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Evaluation will show both overall performance and class-level behavior

[INSERT]

Validation accuracy

Final Colab value

[INSERT]

Validation loss

Final Colab value

[INSERT]

Best accuracy

Final Colab value

[INSERT]

Stopping epoch

Final Colab value

Learning curves

Include accuracy and loss curves to show training progress and validation behavior.

Class-level behavior

Include classification metrics and a confusion matrix to show where activities are distinguished or confused.

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The project produces a reusable inference package



har_lstm_model.keras

Trained Keras model.



har_scaler.pkl

Fitted feature scaler.



har_label_encoder.pkl

Activity label encoder.



har_lstm_predictions.csv

Exported prediction results.

Submission check

Verify exact filenames in the final Colab file browser before submission.

Model · scaler · encoder · predictions

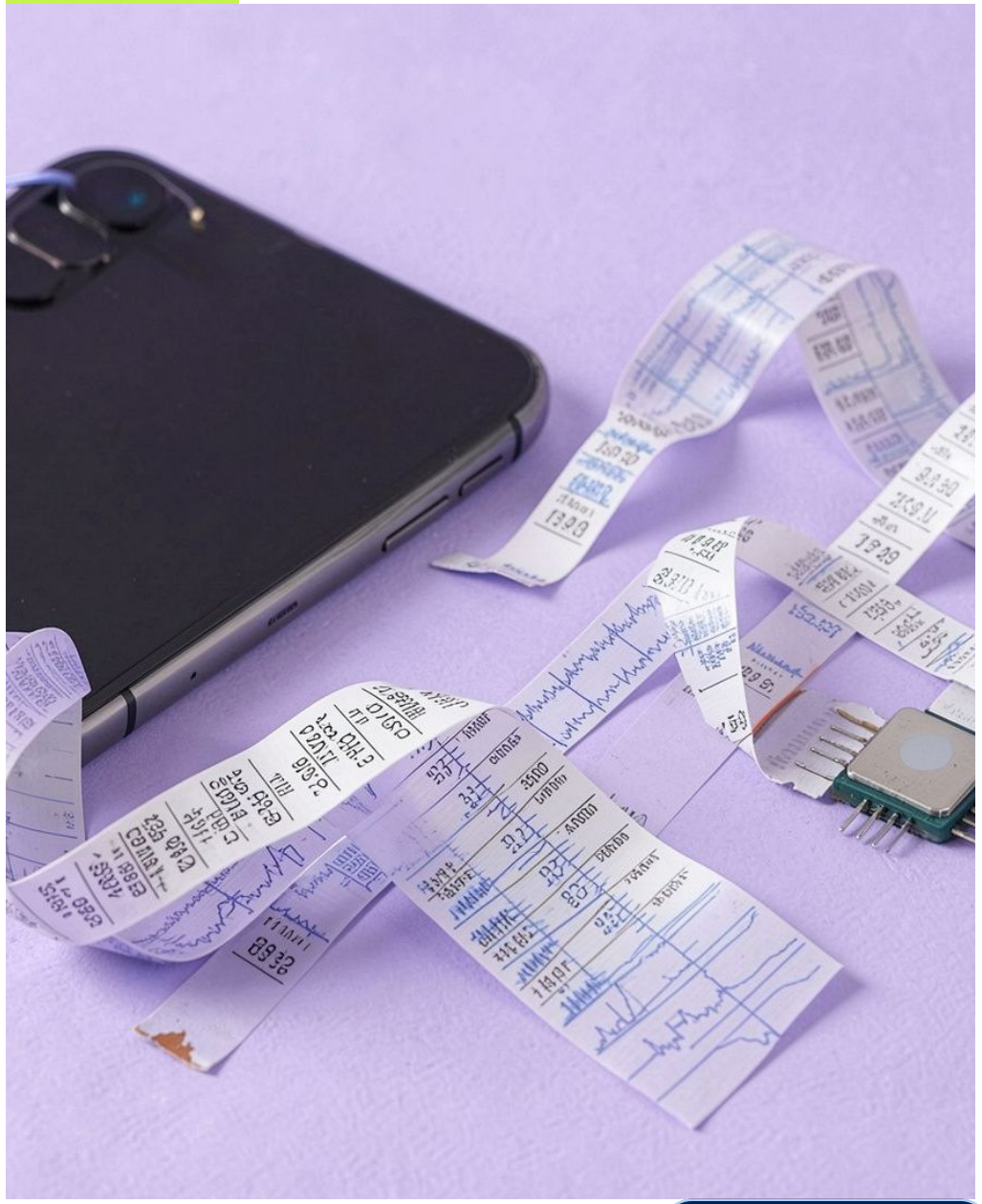
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LIMITATION

The main limitation is sequence representation

A feature vector reshaped to (features, 1) is a sequence by form, not raw time-windowed sensor data.

Performance depends on correct data loading, preprocessing, and label alignment. Results may differ for new users, devices, and environments.



The next step is learning from raw time-windowed

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